

Question

Iron powder reacts with nitric acid of different concentrations at various temperatures, generating multiple nitrogen-containing reduction products, including one salt and several gases. A series of gaseous products obtained from the reaction of iron with excess nitric acid at different temperatures and concentrations were collected. After drying, their density was measured to be $1.545 \times 10^{-3} \text{ g/cm}^3$ under standard conditions. After passing these gases through a NaOH solution, the density changed to $1.416 \times 10^{-3} \text{ g/cm}^3$. Subsequently, after passing through a KMnO_4 solution, the density became $1.429 \times 10^{-3} \text{ g/cm}^3$. (All densities are measured under standard conditions.)

It is known that after each absorption step by the solution, the gases were dried appropriately to remove water vapor without additional gas loss.

Considering only the gaseous products generated from the reaction between iron and nitric acid, determine the coefficient of nitric acid in the overall chemical reaction equation when simplified to the smallest whole number ratio. (Assume that when the ratios of all gases are simplified to the smallest whole numbers, their coefficients are single-digit.)

A. 62

B. 120

C. 132

D. 146

E. 182

Answer

Correct Answer: **E**

Detailed Explanation

The reaction of iron with nitric acid can produce possible products such as NH_4NO_3 , N_2 , NO , N_2O , NO_2 . In the problem, the step involving $NaOH$ solution is where NO and NO_2 are absorbed by $NaOH$, undergoing a comproportionation reaction. Additionally, if NO_2 is in excess compared to NO , the excess NO_2 can also directly react with the $NaOH$ solution and be completely consumed. Therefore, the mixed gas after passing through the $NaOH$ solution cannot contain NO_2 . The change in gas density before and after passing through the $KMnO_4$ solution indicates that the mixed gas before passing through the $KMnO_4$ solution still contains NO , and the oxidation process of NO occurs.

CHECKPOINT

1 PTS

The product gases include N_2 , N_2O , NO , NO_2

The density of $1.545 \times 10^{-3} g/cm^3$ under standard conditions corresponds to a molecular weight of $1.545 \times 10^{-3} \times 22.413 \times 10^3 = 34.63 g/mol$; the density of $1.416 \times 10^{-3} g/cm^3$ corresponds to a molecular weight of $1.416 \times 10^{-3} \times 22.413 \times 10^3 = 31.74 g/mol$; the density of $1.429 \times 10^{-3} g/cm^3$ corresponds to a molecular weight of $1.429 \times 10^{-3} \times 22.413 \times 10^3 = 32.03 g/mol$.

CHECKPOINT

1 PTS

The three densities correspond to molecular weights of 34.62, 31.74, 32.03 g/mol

Assuming the proportion of N_2 is 1, let the ratio of N_2O to N_2 be x . Then the reduced mass of the two can be expressed as:

$$[1 \times 14.01 \times 2 + x \times (14.01 \times 2 + 16.00)] / (1 + x) = 32.03$$

Solving for x gives $x = 0.334 \approx 1/3$

Therefore, $N_2 : N_2O = 3 : 1$

Assuming after passing through the $NaOH$ solution, $NO : N_2 = y$, then the reduced mass of NO , N_2 , N_2O can be expressed as:

$$[3 \times 14.01 \times 2 + 1 \times (14.01 \times 2 + 16.00) + y \times (14.01 + 16.00)] / (3 + 1 + y) = 31.74$$

Solving for y gives $y = 0.64 \approx 2/3$ (the denominator cannot exceed 3, otherwise the coefficient of N_2 would exceed 10)

Assuming $NO_2 : N_2O = z$, then the reduced mass of NO_2, NO, N_2, N_2O can be expressed as:

$$[3 \times 14.01 \times 2 + 1 \times (14.01 \times 2 + 16.00) + (2/3 + z) \times (14.01 + 16.00) + z \times (14.01 + 16.00 \times 2)] / (3 + 1 + 2/3 + 2z) = 34.63$$

Solving for z gives $z = 1.997 \approx 2$

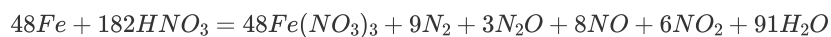
Therefore, $N_2 : N_2O : NO : NO_2 = 3 : 1 : (2/3 + 2) : 2 = 9 : 3 : 8 : 6$

CHECKPOINT

2 PTS



Balancing the equation:



Alternatively, the coefficients can be calculated directly based on the charge required for the redox reaction:

$$9 \times 10 + 3 \times 8 + 8 \times 3 + 6 \times 1 = 144$$

The coefficient for Fe is $144/3 = 48$

The total number of N is $9 \times 2 + 3 \times 2 + 8 + 6 + 48 \times 3 = 182$

Therefore, the coefficient is 182.

CHECKPOINT

2 PTS

The coefficient of HNO_3 is 182